Ordering a PCB through JLCPCB.

These notes attempt to provide a guide to the process of placing an order with **JLCPCB**. It is assumed that the PCB design has been produced using **EAGLE**, but the process is very similar for other packages, just change the design file extension to suit. **JLCPCB** support standard Gerber output (Gerber files in RS-274x format) and Excellon drill output files generated from *most* CAD packages (see https://support.jlcpcb.com/article/22-how-to-generate-the-gerber-files)

For most ECS PCB design work we recommend using **EAGLE CAD** and the corresponding Gerber file generation can be realised by following the instructions on **JLCPCB** : How to export Eagle PCB to gerber files

You'll note that there is a handy **EAGLE CAM** (computer aided manufacturing) file which you can upload from the above link to your working **CAM** directory : either 2-Layer-JLC.cam or 4-Layer-JLC.cam

Running the appropriate CAM job from within EAGLE will auto-generate a zip file containing all the Gerber/Excellon files for **JLCPCB** to build your PCB.

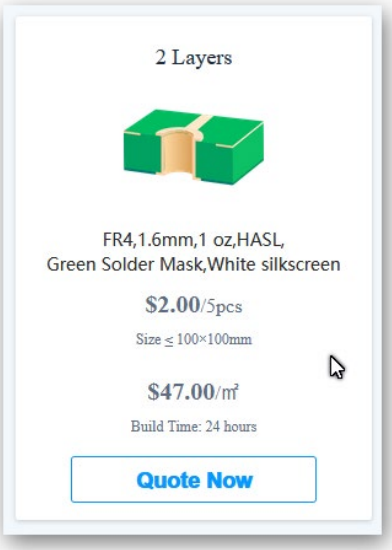
In fact, with this zipped information, many other manufacturers can also quote prices and build your PCB in the same way.

*** ***Rename your zip file*** by prefixing with your UoS login ID ***

e.g. if my zip file was **`CMBoard.zip`**, rename to **`jh17.CMBoard.zip`**. This will help us to identify the owner of the job if ordering things get mixed up.

Once you've generated your zipped manufacturing files, the next step is to obtain an online **quotation**. (*** do not proceed to placing an order as this will be done on your behalf by an ECS buyer)

Proceed to https://jlcpcb.com/ and scroll down to (for a 2-layer board):



Click on '**Quote Now**'

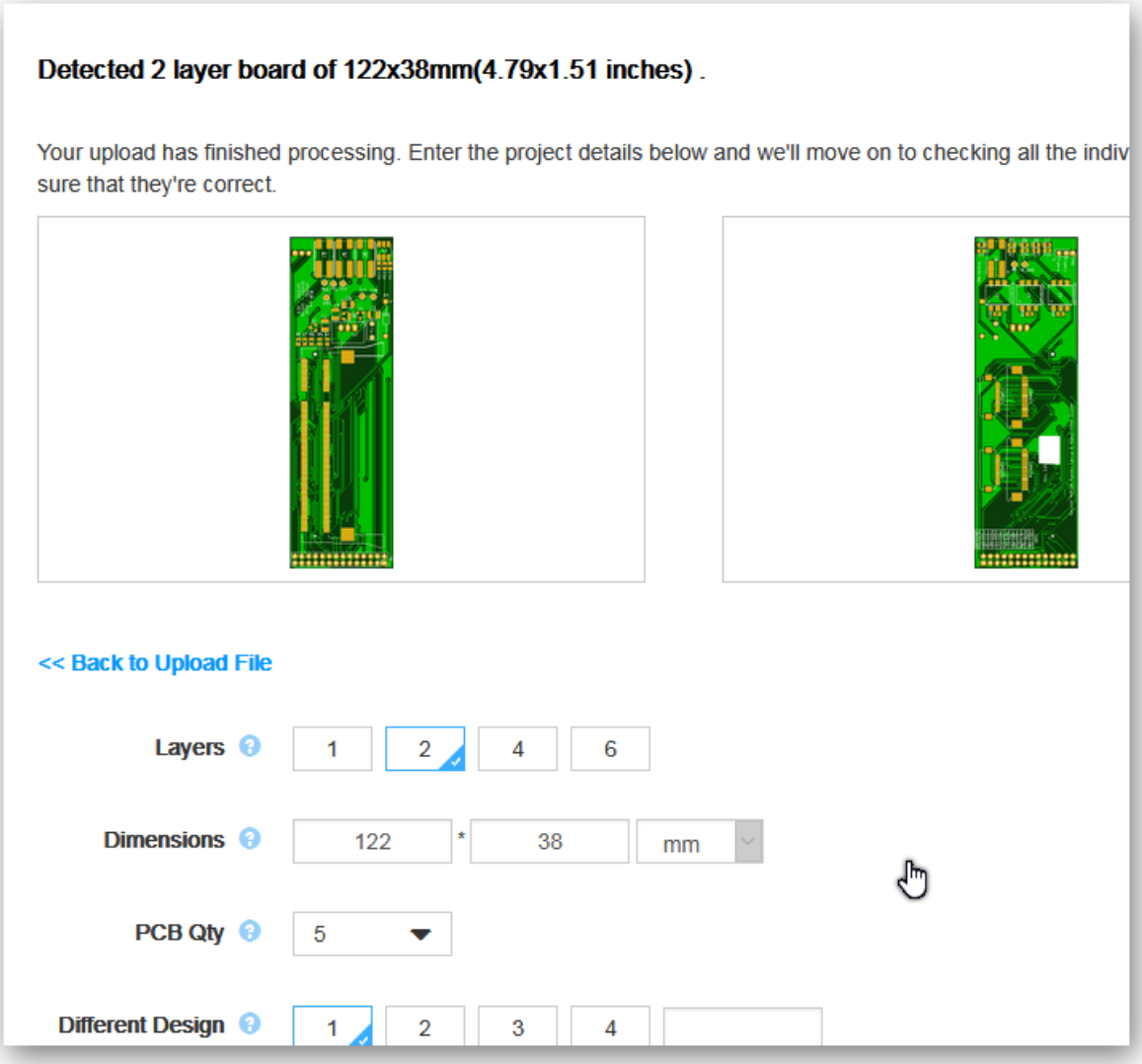
Obtaining a quote:

Drag and drop your zipped manufacturing file onto the ‘Add your gerber file’ block:



After your zip file gets uploaded, you’ll be presented with an updated page which now includes (for 2-layer) a TOP and BOTTOM rendered image of your board.

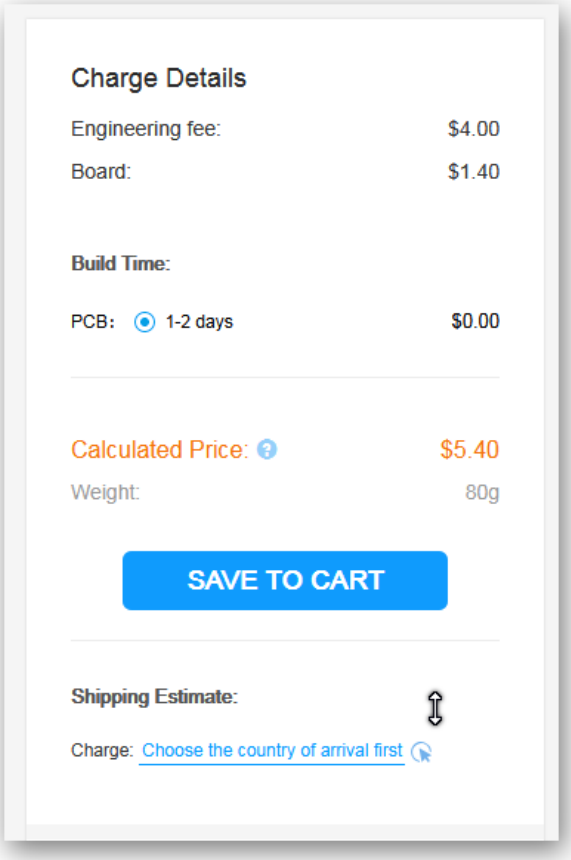
e.g.



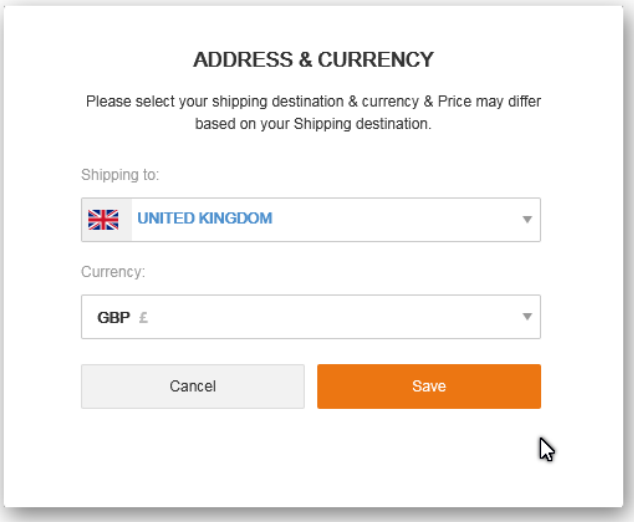
The dimensions will have been auto detected and a default build QTY of 5 is selected. If you need more, update the QTY at this point.

All other defaults will be fine for most purposes, but of course you can change these parameters if required.

You will notice that the costing panel will indicate something like this:

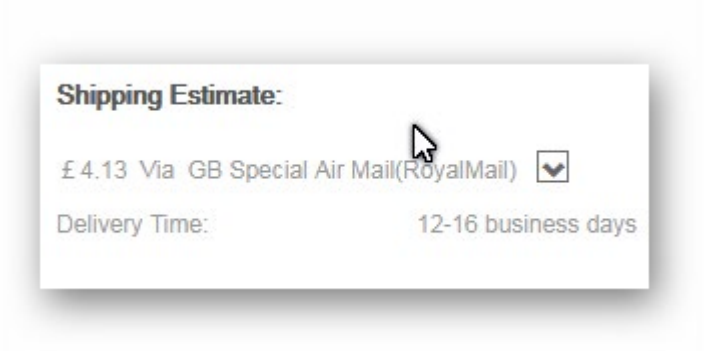


Time to add some shipping information – so change the location & currency:

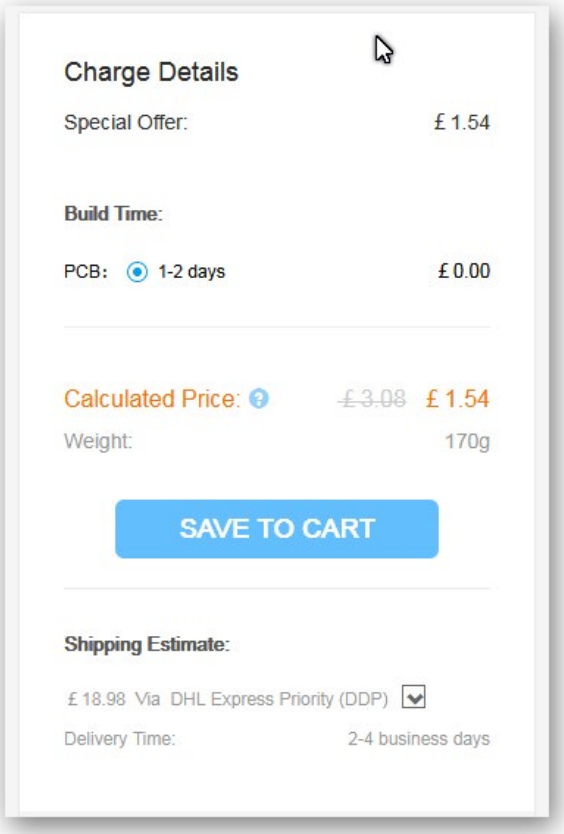


Next, the shipping estimate drop-down defaults to the GB ‘special’ which is cheap but can take a couple of weeks.

This can be changed of course, but the rule-of-thumb is quicker=more £££



So your final Charge Details panel may look like this (note that a ‘special offer’ often gets applied (JLCPCB really want this order!):

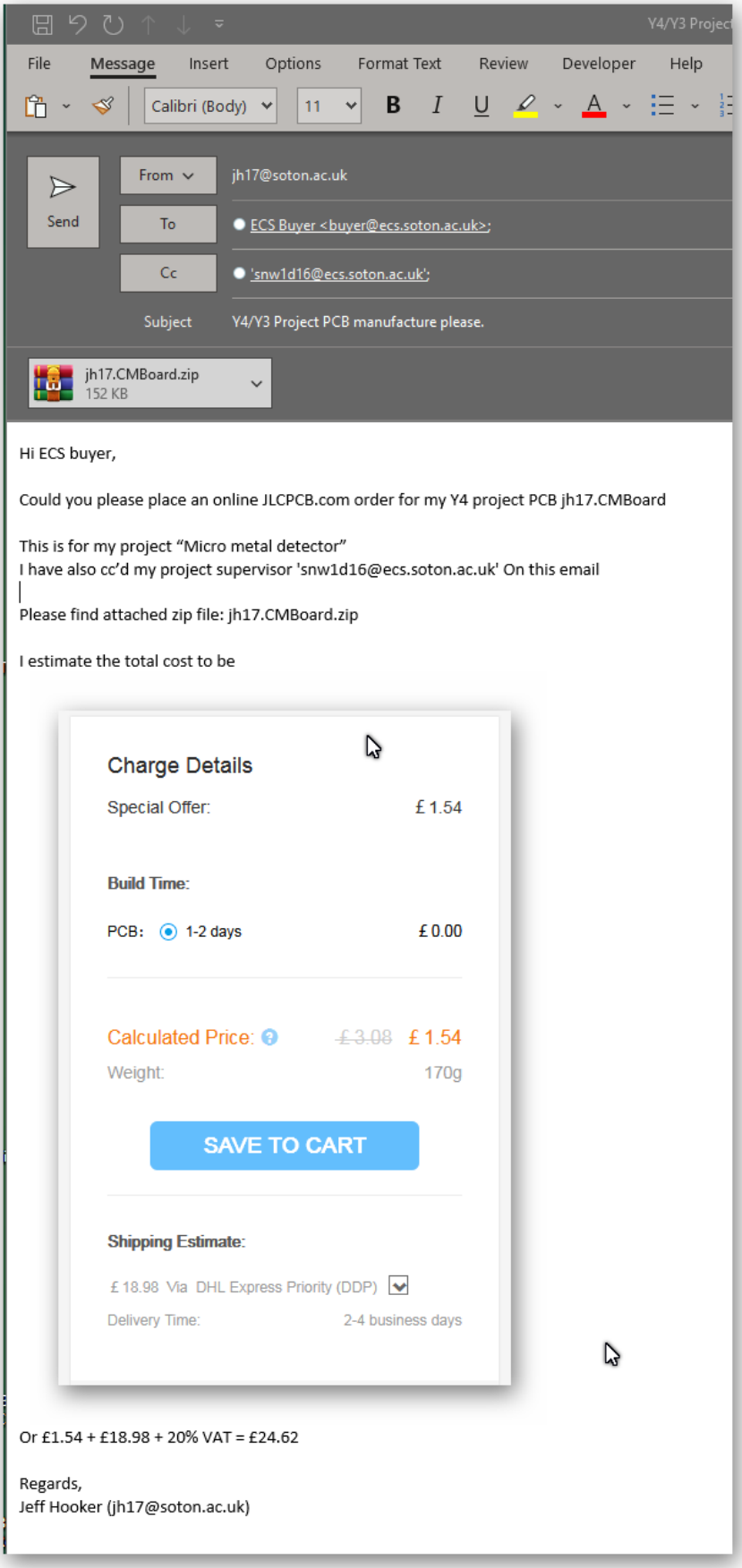


***** DO NOT SAVE TO CART *****

Instead, either screen-shot this panel *or* copy/paste the info within to a new email address:

buyer@ecs.soton.ac.uk This is the university buyer who will purchase the order on your behalf.

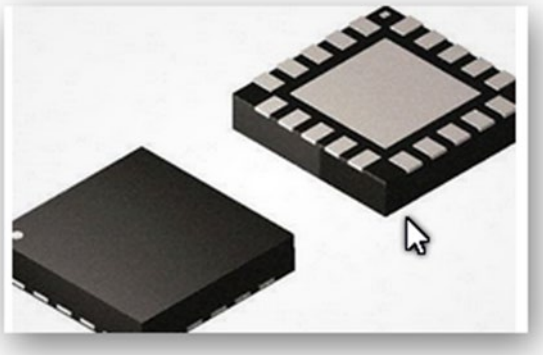
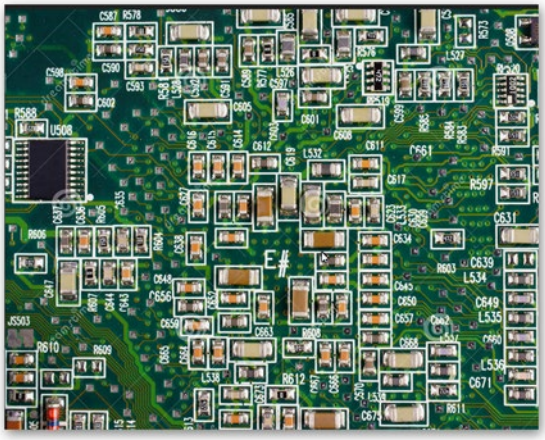
Your email request to buyer@ecs.soton.ac.uk will look something like this:



When your PCB package is delivered it will arrive at B16.1013 (within main L1 lab B16.1003) and the technicians there will email you so that you can arrange to pick it up.

Note regarding solder-paste stencils.

If you design includes fine-pitch SMD packages (especially QFN packages) or perhaps a high number of passive SMDs:

	
QFN Package	Large number of passive SMDs

Then you will find that ordering a solder-paste stencil is pretty much essential. This allows for precise depositing of solder-paste onto pads so that your hand-placement of devices is greatly sped up.

A populated board can then be soldered in a matter of minutes using a reflow oven (available in B16 projects lab)

To order a stencil at the same time as ordering a PCB is very easy.

At the above '**obtaining a quote**' stage you will notice that you have the option:



Toggle this to 'on'. More stencil options appear:

Framework : **NO**
Dimensions : (a default value appears here)
Customized size : select **NO** (*read on)
Stencil Side : select '**TOP & BOTTOM (on separate stencils**)
Or if *all* of your SMDs are on one side select either '**TOP'** or '**BOTTOM'**
You may also choose just one side if the other side is easily hand-solderable.
Stencil Qty : 1
Electropolishing : **NO**
Fiducials : **No Fiducial**